TMHM: IA 1

1. Explain Circulatory Disorders

Answer

The **circulatory or cardiovascular system** consists mainly of the **heart, blood vessels and blood**. The heart acts as a pump, while arteries, veins and capillaries transport blood throughout the body. The system transports oxygen, nutrients, hormones and waste products and also helps in regulation and protection.

Major Circulatory Disorders

1. Hypertension

Hypertension is the **persistent elevation of blood pressure**.

- It causes increased pressure on the walls of blood vessels.
- Persistent high blood pressure can affect the normal functioning of the cardiovascular system.
- It is associated with other cardiovascular complications.

2. Atherosclerosis

Atherosclerosis is the **hardening and narrowing of arteries due to cholesterol deposits**.

- Cholesterol deposits accumulate in arterial walls.
- The arterial lumen becomes narrow.
- Blood flow to tissues is reduced.

3. Heart Attack / Myocardial Infarction

A myocardial infarction occurs due to **blockage of a coronary artery**.

Blockage of coronary artery → reduced blood supply → death of heart tissue

The affected heart muscle does not receive adequate oxygen and nutrients.

4. Stroke

Stroke occurs due to **blockage or rupture of blood vessels supplying the brain**.

- Blockage prevents adequate blood supply to brain tissue.
- Rupture causes bleeding within or around the brain.
- Brain tissue can consequently be damaged.

5. Anemia

Anemia is characterized by a **low hemoglobin level or reduced red blood cell count**.

Since RBCs carry oxygen using hemoglobin, reduction in their number or hemoglobin reduces oxygen transport.

6. Leukemia

Leukemia is a **cancer of white blood cells (WBCs)**.

It affects the normal blood-forming system and consequently interferes with normal blood functions.

7. Varicose Veins

Varicose veins are **swollen veins caused by failure of venous valves**.

- Valves normally prevent backward flow of blood.
- Valve failure allows blood to accumulate in veins.
- The veins become enlarged and swollen.

8. Shock

Shock is an **acute circulatory failure resulting in inadequate tissue perfusion and oxygenation**.

Major types include:

- **Hypovolemic shock** – blood/fluid loss.
- **Cardiogenic shock** – failure of the heart as a pump.
- **Septic shock** – severe infection causing vasodilation and capillary leakage.

- **Anaphylactic shock** – severe allergic reaction.
- **Neurogenic shock** – loss of sympathetic tone due to spinal cord injury or anesthesia.

Common features include hypotension, rapid weak pulse, altered consciousness and reduced urine output.

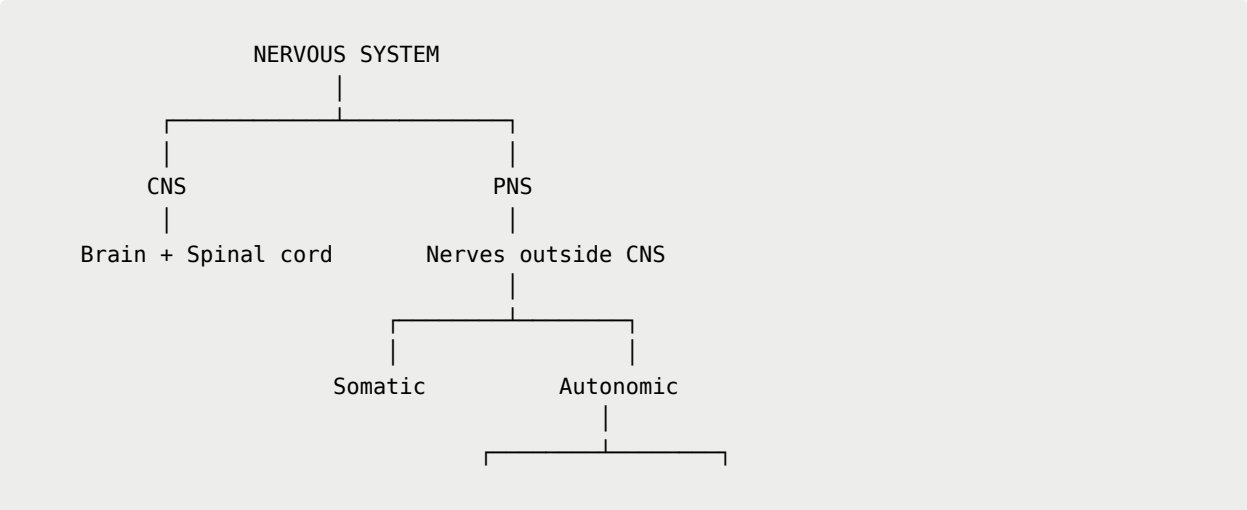
Functions of Circulatory System

1. Transport of oxygen to tissues.
2. Transport of carbon dioxide to lungs.
3. Transport of nutrients to cells.
4. Transport of hormones to target organs.
5. Transport of waste products to kidneys, lungs and skin.
6. Regulation of temperature, pH and fluid balance.
7. Protection through WBCs, platelets and antibodies.

2. Explain Classification of Nervous System

Answer

The **nervous system** is responsible for receiving information, processing it and transmitting appropriate responses throughout the body. It is mainly divided into:



```
                                    |                 |
                              Sympathetic      Parasympathetic
```

1. Central Nervous System (CNS)

The CNS consists of:

- **Brain**
- **Spinal cord**

Brain

The brain acts as the **command center**.

Functions:

- Receives sensory information.
- Processes and integrates information.
- Generates motor responses.

Spinal Cord

The spinal cord acts mainly as a **relay center**.

- Transmits information between the brain and peripheral nerves.
- Participates in transmission of sensory and motor information.

2. Peripheral Nervous System (PNS)

The PNS consists of **all nerves outside the brain and spinal cord**.

Functions:

- Carries sensory information from the body to the CNS.
- Carries commands from the CNS to different parts of the body.

The PNS is further divided into:

A. Somatic Nervous System

The somatic nervous system controls **voluntary movements**.

It is involved in conscious control of skeletal muscles and transmission of sensory information.

B. Autonomic Nervous System

The autonomic nervous system controls **involuntary functions**.

Examples include:

- Heart rate
- Digestion
- Respiration

It is further associated with sympathetic and parasympathetic control of involuntary activities.

Difference between CNS and PNS

CNS	PNS
Brain and spinal cord	Nerves outside CNS
Processing and integration	Transmission of information
Acts as command/relay center	Connects CNS with body
Generates appropriate responses	Carries sensory and motor signals

Thus, the nervous system provides coordination between sensory input and appropriate responses and helps maintain normal body functions.

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3. Explain Carbohydrate Glycolysis Metabolism

Answer

Carbohydrate metabolism refers to the chemical processes involved in the breakdown of carbohydrates to produce energy and the synthesis/storage of carbohydrates required by the body. The primary pathway is **glycolysis**.

Glycolysis

Glycolysis is the metabolic pathway in which **one molecule of glucose containing six carbon atoms is converted into two molecules of pyruvate, each containing three carbon atoms**.

Location

Glycolysis occurs in the **cytoplasm of the cell**.

Overall pathway

```
            GLUCOSE
               ↓
      Glucose-6-phosphate
               ↓
  Fructose-1,6-bisphosphate
               ↓
Glyceraldehyde-3-phosphate
               ↓
            PYRUVATE
```

The major sequence is:

Glucose → Glucose-6-phosphate → Fructose-1,6-bisphosphate → Glyceraldehyde-3-phosphate → Pyruvate

Energy Yield

For one molecule of glucose:

- **Net ATP = 2 ATP**
- **NADH = 2 molecules**

Thus, glycolysis provides a rapid source of energy.

Aerobic and Anaerobic Conditions

Glycolysis can occur under both aerobic and anaerobic conditions.

Aerobic condition

```
Glucose
   ↓
Glycolysis
   ↓
Pyruvate
   ↓
Citric Acid Cycle
   ↓
Electron Transport Chain
   ↓
Large amount of ATP
```

The pyruvate is processed through the **citric acid/Krebs cycle** in mitochondria, producing NADH and $FADH_2$. The electron transport chain uses these electron carriers to produce large amounts of ATP through oxidative phosphorylation.

Anaerobic condition

When oxygen is insufficient, such as during intense muscle activity:

```
Glucose → Glycolysis → Pyruvate → Lactate
```

Pyruvate is converted into **lactate**.

Other Carbohydrate Metabolic Pathways

Glycogenesis

Excess glucose is converted into **glycogen** for storage in the liver and skeletal muscles.

Glucose → Glucose-6-phosphate → Glucose-1-phosphate → UDP-glucose → Glycogen

Glycogenolysis

Stored glycogen is broken down into glucose-1-phosphate and subsequently glucose-6-phosphate.

- Liver: helps maintain blood glucose.
- Muscle: provides glucose-6-phosphate for ATP production.

Gluconeogenesis

Glucose is synthesized from non-carbohydrate sources such as:

- Amino acids
- Glycerol
- Lactate

It occurs mainly in the liver during fasting or starvation.

Importance of Carbohydrates

1. Major dietary source of energy.
2. Provide **4 kcal/g**.
3. Serve as precursors for fats and amino acids.
4. Form structural components.
5. Glycogen serves as an energy storage form.
6. Ribose and deoxyribose are components of RNA and DNA.
7. Provide energy for the brain.
8. Dietary fibre helps prevent constipation.

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4. Explain Cell Functions and Any Five Important Cell Organelles

Answer

A **cell is the smallest structural and functional unit of a living organism**. Cells may be prokaryotic or eukaryotic. Eukaryotic cells contain a nucleus and membrane-bound organelles.

Functions of Cells

1. Provide structure and support.
2. Carry out metabolism such as respiration and protein/lipid synthesis.
3. Participate in growth and reproduction through mitosis and meiosis.
4. Communicate through signaling molecules.
5. Maintain **homeostasis** within the cell.

Five Important Cell Organelles

1. Nucleus

The nucleus is the **control center of the cell**.

- Surrounded by a nuclear membrane.
- Contains DNA/genetic material.
- Controls important cellular activities.
- Involved in cell division and protein synthesis.

2. Mitochondria

Mitochondria are called the **“powerhouse of the cell.”**

- Produce energy in the form of **ATP**.
- Important for cellular energy production.

3. Endoplasmic Reticulum (ER)

ER is an internal membrane system.

Rough ER:

- Contains ribosomes.
- Mainly involved in protein synthesis.

Smooth ER:

- Involved in lipid synthesis.

- Helps in detoxification.

4. Golgi Apparatus

The Golgi apparatus:

- Modifies proteins.
- Stores proteins.
- Packages proteins for transport.

5. Lysosomes

Lysosomes contain **digestive enzymes**.

- Digest cellular materials.
- Break down unwanted materials and cellular components.
- They are sometimes called **“suicidal bags”** of the cell.

Other Important Structures

Cell membrane: Thin, flexible, semi-permeable boundary that controls movement of substances into and out of the cell.

Cytoplasm: Gel-like material containing organelles where many metabolic activities occur.

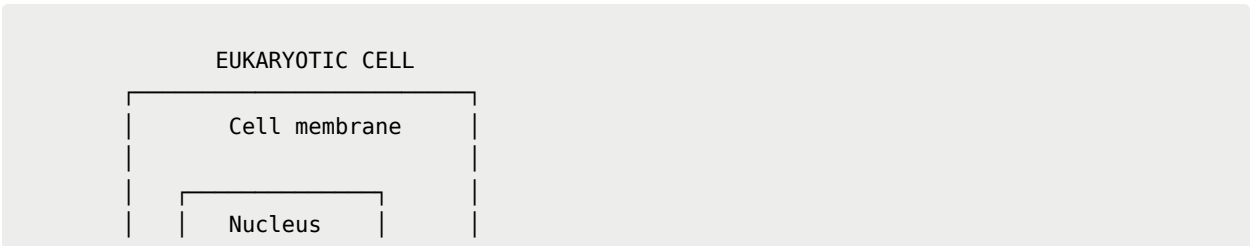
Ribosomes: Sites of protein synthesis.

Vacuoles: Storage structures for water, food and waste.

Chloroplasts: Present in plants and carry out photosynthesis.

The major organelles and their functions are summarized in the cell structure material.

Simple Diagram



```
|   └──────────────┘      |
|                         |
|  Mitochondria           |
|  Rough ER               |
|  Smooth ER              |
|  Golgi apparatus        |
|  Lysosomes              |
|  Ribosomes              |
|                         |
└─────────────────────────┘
```

5. Explain Micro and Macronutrients

Answer

Nutrients are substances required by the body for **energy production, growth, maintenance and normal physiological functions**.

The food constituents are broadly divided into **major and minor food components**. Major components include water, protein, carbohydrates and fats, while minor components include vitamins and minerals along with other food constituents.

Macronutrients

Macronutrients are nutrients required in relatively larger amounts.

The important macronutrients are:

1. Carbohydrates

- Major dietary source of energy.
- Provide **4 kcal/g**.
- Glucose is used by cells as an energy source.
- Excess glucose can be stored as glycogen.
- Also participate in structural and cellular functions.

2. Proteins

Proteins are nitrogen-containing organic macromolecules made up of amino acids.

Functions:

- Structural support.
- Enzymatic activity.
- Hormonal functions.
- Transport.
- Blood clotting.
- Immune functions.
- Muscle contraction.

Proteins perform both structural and dynamic functions in living organisms.

3. Fats/Lipids

Lipids are important energy-rich molecules.

Functions:

- Efficient energy source.
- Energy storage in adipose tissue.
- Thermal insulation.
- Structural components of cell membranes.
- Formation of hormones.
- Source of fat-soluble vitamins A, D, E and K.

Fats provide approximately **9 kcal/g**.

4. Water

Water is listed among the major food components.

It is essential for normal physiological processes and forms a major component of the body.

Micronutrients

Micronutrients are required in **small amounts** but are important for metabolic processes and normal functioning.

The major micronutrients are:

1. Vitamins

Vitamins are required in small quantities for various metabolic processes.

There are **13 essential vitamins**, divided into:

Fat-soluble vitamins:

- Vitamin A
- Vitamin D
- Vitamin E
- Vitamin K

Water-soluble vitamins:

- Eight B-complex vitamins
- Vitamin C

Example: Vitamin A supports:

- Immune function
- Healthy skin
- Reproduction and growth
- Vision

2. Minerals

Minerals are also important micronutrients required for normal body functions and metabolic processes.

Difference

Macronutrients	Micronutrients
Required in larger amounts	Required in smaller amounts

Macronutrients	Micronutrients
Mainly carbohydrates, proteins and fats	Mainly vitamins and minerals
Important sources of energy and structural materials	Important for metabolic and physiological processes
Carbohydrates and proteins: 4 kcal/g	Generally do not provide energy

6. Explain Etiology – Mechanism and Major Causes

Answer

Etiology

Etiology refers to the **cause or causes of a disease**. Disease may result from genetic, environmental, nutritional, immunological, lifestyle or age-related factors.

Major Causes of Disease

1. Genetic Causes

Diseases may occur due to abnormalities in **DNA or chromosomes**.

Mechanism

```
Mutation
   ↓
Abnormal protein/enzyme
   ↓
Loss of normal cell function
   ↓
Structural/functional abnormality
   ↓
Disease
```

Examples:

- Sickle cell anemia

- Down syndrome
- Cystic fibrosis

2. Environmental Causes

Physical agents

- Trauma → fractures and wounds.
- Radiation → DNA mutations and cancers.
- Extreme temperatures → burns and frostbite.

Chemical agents

- Poisons such as arsenic and lead damage tissues.
- Drugs and alcohol may cause liver and kidney damage.

Biological agents

- Bacteria → TB, cholera.
- Viruses → HIV, influenza.
- Fungi → ringworm.
- Parasites → malaria.

3. Nutritional Causes

Deficiency

Lack of vitamins, minerals or proteins can cause disease.

Examples:

- Vitamin D deficiency → rickets.
- Vitamin C deficiency → scurvy.
- Iron deficiency → anemia.

Excess

- Overeating → obesity and diabetes.
- Excess vitamin A/D → toxicity.

4. Immunological Causes

Hypersensitivity

Overreaction of the immune system to harmless substances.

Example: asthma and hay fever.

Autoimmunity

Immune system attacks the body's own tissues.

Examples:

- Rheumatoid arthritis
- Lupus

Immunodeficiency

Weak or absent immune system leading to frequent infections.

Example: HIV/AIDS.

5. Lifestyle Causes

- Smoking → lung cancer, COPD and heart disease.
- Alcohol → liver cirrhosis, brain damage and cancer.
- Poor diet/lack of exercise → obesity, diabetes and hypertension.
- Stress → increased risk of depression, heart disease and ulcers.

6. Aging

Aging involves gradual degeneration of cells and tissues.

Mechanisms include:

- Reduced repair capacity.

- Increased oxidative stress.
- Progressive cellular deterioration.

Examples:

- Osteoporosis
- Alzheimer's disease
- Parkinson's disease

Conclusion

Disease generally results from interaction between causative factors and the body's cellular and tissue responses. The same causative factor may produce different effects depending on the individual and circumstances.

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7. Explain Pathogenesis

Answer

Pathogenesis refers to the **development of a disease and the sequence of events through which the disease develops**.

Disease development occurs in stages, and the exact course depends on the individual, causative factor and medical intervention.

General Sequence

```
CAUSE / ETIOLOGICAL AGENT
          ↓
      CELLULAR EFFECT
          ↓
     TISSUE RESPONSE
          ↓
    FUNCTIONAL CHANGE
          ↓
   SIGNS AND SYMPTOMS
          ↓
     COMPLICATIONS
```

```
        ↓
OUTCOME / RECOVERY
```

Stages

1. Exposure / Initiation

The disease begins when the body is exposed to a causative factor such as:

- Genetic abnormality
- Microorganism
- Chemical agent
- Physical injury
- Nutritional deficiency
- Immune abnormality

2. Cellular and Tissue Changes

The causative factor produces changes in cells and tissues.

For example, in infectious disease:

Entry of pathogen → multiplication → tissue damage

Microorganisms may produce toxins or enzymes that damage cells.

3. Host Response

The body responds to the disease-producing factor.

In infection, immune responses may result in:

- Fever
- Inflammation
- Pus formation

4. Development of Clinical Disease

Structural and functional changes become sufficiently significant to produce recognizable **signs and symptoms**.

A collection of signs and symptoms characterizing a disorder is called a **syndrome**.

5. Course of Disease

Depending on the disease, the course may include:

- **Incubation period**
- **Period of full-blown symptoms**
- **Remission**
- **Convalescence**

Example: Infectious Disease

```
Pathogen enters host
       ↓
Colonization/multiplication
       ↓
Toxin or enzyme production
       ↓
Cell/tissue damage
       ↓
Immune response
       ↓
Inflammation and symptoms
       ↓
Recovery / complications
```

Factors Affecting Pathogenesis

1. Type of causative agent.
2. Type of affected cells/tissues.
3. Duration of exposure.
4. Severity of injury.
5. Individual patient characteristics.
6. Medical intervention.

Thus, pathogenesis explains **how a disease develops from its cause to its clinical manifestations and final outcome**.

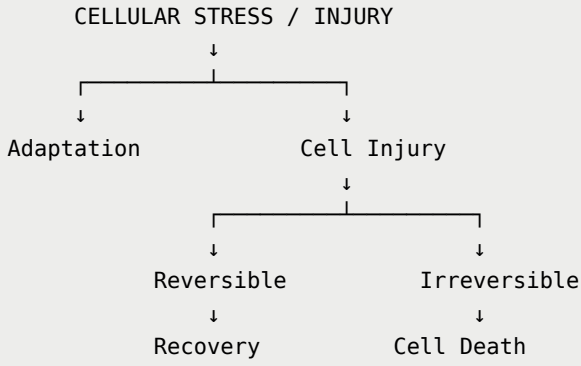
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8. Explain Cellular Response to Injury

Answer

Cells are continuously exposed to stress, injury and environmental changes. The cellular response depends on the **type of cell, type of injury, duration and severity**.

The major outcomes are:



1. Cellular Adaptation

When exposed to stress, cells may adapt in order to survive.

A. Atrophy

Decrease in **cell size**.

Example: muscle wasting after immobilization.

B. Hypertrophy

Increase in **cell size**.

Example: cardiac hypertrophy in hypertension.

C. Hyperplasia

Increase in **cell number**.

Example: endometrial hyperplasia.

D. Metaplasia

Replacement of one mature cell type by another.

Example: in smokers, respiratory epithelium changes from columnar to squamous epithelium.

E. Dysplasia

Abnormal cell growth and differentiation, often considered precancerous.

2. Reversible Cell Injury

Reversible injury is an **early and mild stage** in which the cell can return to normal if the damaging stimulus is removed.

Features

- Cell swelling.
- Fatty changes.
- Membrane blebbing.

3. Irreversible Cell Injury

Irreversible injury occurs when damage is **severe and persistent**.

It ultimately results in cell death.

Features

- Mitochondrial damage.
- Membrane rupture.
- Nuclear changes.

4. Cell Death

A. Necrosis

Necrosis is **uncontrolled pathological cell death**.

Types include:

- Coagulative necrosis → myocardial infarction.
- Liquefactive necrosis → brain abscess, stroke.
- Caseous necrosis → tuberculosis.
- Fat necrosis → acute pancreatitis.
- Gangrenous necrosis → limb ischemia.

B. Apoptosis

Apoptosis is **programmed cell death**.

It may be physiological or pathological.

Functions:

- Embryogenesis.
- Elimination of self-reactive lymphocytes.
- Removal of old or damaged cells.

Features:

- Cell shrinkage.
- Chromatin condensation.
- Formation of apoptotic bodies.
- Usually no inflammation.

5. Autophagy

Autophagy is a **“self-eating” process** in which the cell digests its own organelles, particularly during starvation.

Thus, cellular responses to injury range from adaptation and reversible injury to irreversible injury and cell death.

9. Explain Secondary Plant Metabolites and Their Biological Functions

Answer

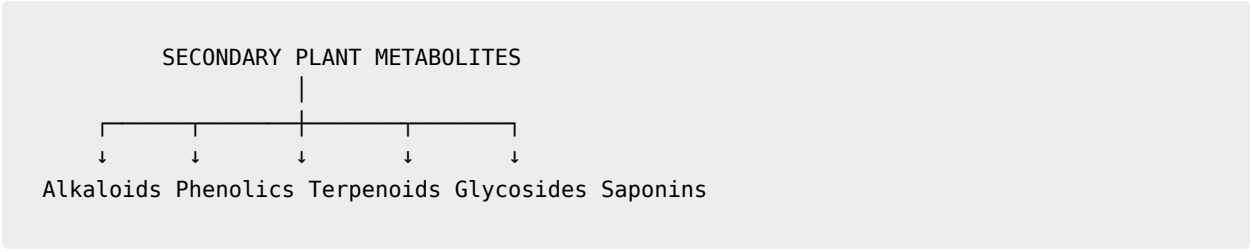
Secondary plant metabolites, also called **secondary plant chemicals**, are organic compounds produced by plants that are not directly involved in basic growth, development or reproduction.

They are important in:

- Plant defense.
- Signaling.
- Ecological interactions.
- Medicine.
- Agriculture.
- Industry.

They act as a plant's **defense and survival system**.

Major Classes



1. Alkaloids

Alkaloids are **nitrogen-containing natural compounds**, usually having nitrogen within a ring system.

Properties

- Generally basic.

- Many are crystalline solids.
- Some are liquids such as nicotine.
- Often bitter.
- May be toxic at high doses.

Biological/Medicinal Functions

- Plant protective substances.
- Metabolic regulators.
- Nitrogen reservoirs.
- Pharmacological effects in humans and animals.

Examples

- **Morphine** → opioid analgesic.
- **Quinine** → antimalarial.
- **Caffeine** → central nervous stimulant.
- **Nicotine** → neurotoxin and natural insecticide.

2. Terpenoids

Terpenoids are natural organic compounds built from **5-carbon isoprene units**.

Properties

- Often volatile.
- Hydrophobic.
- Strongly aromatic.
- Contribute to plant scent, colour and flavour.

Functions and Examples

- **Menthol** → cooling sensation and insect repellent.
- **β-carotene** → pigment and vitamin A precursor.

- **Taxol** → anticancer agent.
- **Rubber** → elasticity and commercial use.

3. Glycosides

Glycosides consist of:

Sugar part (glycone) + non-sugar part (aglycone)

They are joined by a **glycosidic bond**.

Major types:

Cardiac glycosides

Strengthen heart muscle contraction and slow heart rate.

Example: **Digitalis**.

Cyanogenic glycosides

Release toxic **hydrogen cyanide (HCN)** when enzymatically broken down.

Examples: stone fruits, cassava and bitter almond.

Anthraquinone glycosides

Have stimulant laxative action.

Examples: **Senna and aloe vera**.

4. Saponins

Saponins are plant glycosides that produce a **soap-like lather in water**.

Properties

- Amphiphilic.
- Form micelles in water.
- May have hemolytic activity.

Biological Functions

- Antifungal and antimicrobial activity.
- Reduce cholesterol by binding bile acids.
- Provide plant defense against insects.

Sources include soybean, quinoa, ginseng and fenugreek.

5. Phenolics

Phenolics contain at least one **aromatic benzene ring directly bonded to one or more hydroxyl groups**.

Properties

- Antioxidant activity.
- Scavenge free radicals.
- Absorb UV light.
- Tannins can bind and precipitate proteins.

Biological Functions

- Protect plants against herbivores.
- Protect against fungal pathogens and insects.
- Protect against oxidative stress.
- Anthocyanins provide red, purple and blue colours.
- Associated with anti-inflammatory, antimicrobial and cardiovascular protective effects.

Conclusion

Secondary metabolites are not primarily required for basic growth, but they are highly important for **plant defense, adaptation and survival**, and many have significant medicinal and pharmacological applications.

---

10. Explain ADME

Answer

ADME represents the four major processes involved in the movement and fate of a medicinal substance/drug in the body:

A – Absorption
D – Distribution
M – Metabolism
E – Excretion

```
          DRUG ADMINISTRATION
                  ↓
            ABSORPTION
                  ↓
          BLOOD CIRCULATION
                  ↓
           DISTRIBUTION
                  ↓
             TISSUES
                  ↓
            METABOLISM
                  ↓
             EXCRETION
```

The module describes the pharmacological dynamics of medicinal substances through absorption, distribution, metabolism and excretion.

1. Absorption

Absorption is the process by which a drug enters the bloodstream from its site of administration.

Routes

- Oral
- Intravenous
- Intramuscular
- Subcutaneous
- Transdermal

Mechanisms

1. **Passive diffusion** – movement from high to low concentration.
2. **Facilitated diffusion** – carrier-mediated and requires no energy.
3. **Active transport** – energy-dependent.
4. **Endocytosis/exocytosis** – mainly for macromolecules.

Factors affecting absorption

- Route of administration.
- Lipid solubility.
- Molecular size.
- Ionization.
- Formulation.
- Gastrointestinal pH.
- Food.
- Gastric motility.
- Blood flow.

Lipid-soluble drugs generally cross cell membranes more easily.

Bioavailability

Bioavailability is the **fraction of administered drug reaching systemic circulation unchanged**.

- IV route → 100% bioavailability.
- Oral drugs may undergo first-pass metabolism in the liver.

---

2. Distribution

Distribution is the **reversible transfer of a drug from the bloodstream to tissues and organs**.

Factors affecting distribution

- Blood flow to tissues.
- Plasma protein binding.
- Tissue binding.
- Lipid solubility.

Drugs bound to plasma proteins such as albumin are considered inactive, while the free drug is active.

Volume of Distribution

Volume of distribution (Vd) indicates the extent to which a drug is distributed throughout the body.

- High Vd → widely distributed in tissues.
- Low Vd → mainly confined to plasma.

Examples:

- Chloroquine → high Vd.
- Heparin → low Vd.

---

3. Metabolism

Drug metabolism, or **biotransformation**, is the enzymatic conversion of drugs into more water-soluble compounds so that they can be eliminated.

Main Site

The **liver** is the primary site.

The major enzyme system involved is the **Cytochrome P450 (CYP) system**.

Phase I

Functionalization reactions include:

- Oxidation
- Reduction

- Hydrolysis

These reactions introduce or expose functional groups such as:

- –OH
- $–NH_2$
- –SH

Phase II

Phase II consists of **conjugation reactions**.

The drug or metabolite combines with endogenous substances such as:

- Glucuronic acid
- Sulfate
- Glycine

This generally produces more polar compounds that are easier to excrete.

Factors affecting metabolism

- Genetic factors.
- Age.
- Liver disease.
- Enzyme induction/inhibition.
- Diet.
- Environmental factors.

---

4. Excretion

Excretion is the process by which the body **eliminates drugs or their metabolites**.

Major Routes

A. Renal Excretion

The kidney is the primary route.

It involves:

1. **Glomerular filtration**
    - Free/unbound drug passes through glomeruli.
2. **Tubular secretion**
    - Active transport of certain drugs.
3. **Tubular reabsorption**
    - Lipid-soluble drugs may diffuse back into circulation.

B. Biliary Excretion

Large polar conjugates are excreted through:

Bile → intestine → feces

C. Pulmonary Excretion

Volatile substances such as anesthetic gases can be eliminated through the lungs.

Summary of ADME

Process	Meaning	Major point
Absorption	Drug enters blood	Affected by route and physicochemical properties
Distribution	Drug moves from blood to tissues	Affected by blood flow, protein binding and lipid solubility
Metabolism	Drug is chemically modified	Mainly occurs in liver
Excretion	Drug/metabolites leave body	Mainly through kidneys